



E-Commerce Sales Analytics Dashboard

Full Project Report · Visual Screenshots · Data Explanation

Data Period: January 2022 – December 2024

Total Orders: 3,500 · Products: 10 · Regions: 4

Total Revenue: \$10.67M · Profit Margin: 17.3%

Built with React 18 · Recharts · Vite

Visualised with Matplotlib · Report by ReportLab

Prepared by SalesIQ Analytics Engine | May 2026

1. Executive Summary

This report presents a comprehensive analysis of e-commerce sales data spanning three full fiscal years (2022–2024), covering 3,500 transactions across 10 product lines, 3 categories, and 4 geographic regions. The SalesIQ dashboard was built as an interactive React web application and this document provides a full visual walkthrough of every dashboard section, chart, and data insight discovered.

Key Performance Indicators

TOTAL REVENUE \$10.67M All regions · 3 years	TOTAL PROFIT \$1.84M Net after cost	TOTAL ORDERS 3,500 Jan 2022 – Dec 2024	AVG ORDER VALUE \$3,047 Per transaction	PROFIT MARGIN 17.3% Revenue efficiency
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Electronics dominated revenue with \$5.33M (49.9% of total), followed by Accessories at \$3.95M (39.8%) and Office at \$1.09M (10.3%). The West region led geographically with \$2.84M in sales. Camera was the top-grossing individual product at \$1.18M. Profit margins remained consistent across all years at approximately 17%, demonstrating stable operational efficiency.

Project Objectives

- Analyse 3 years of e-commerce sales performance across products, categories, and regions.
- Identify top-performing products and lowest-margin items.
- Uncover seasonal sales trends and year-over-year growth patterns.
- Provide an interactive dashboard for stakeholders to filter and explore data.
- Deliver a GitHub-ready React project for deployment and sharing.

2. Dashboard Overview

The SalesIQ dashboard is a dark-themed, single-page React application featuring four navigable tabs (Overview, Products, Regions, Trends), a year filter, and live-updating Recharts visualisations. All data is embedded in the component — no backend or API is required.

2.1 Combined Monthly Sales & Profit — Screenshot

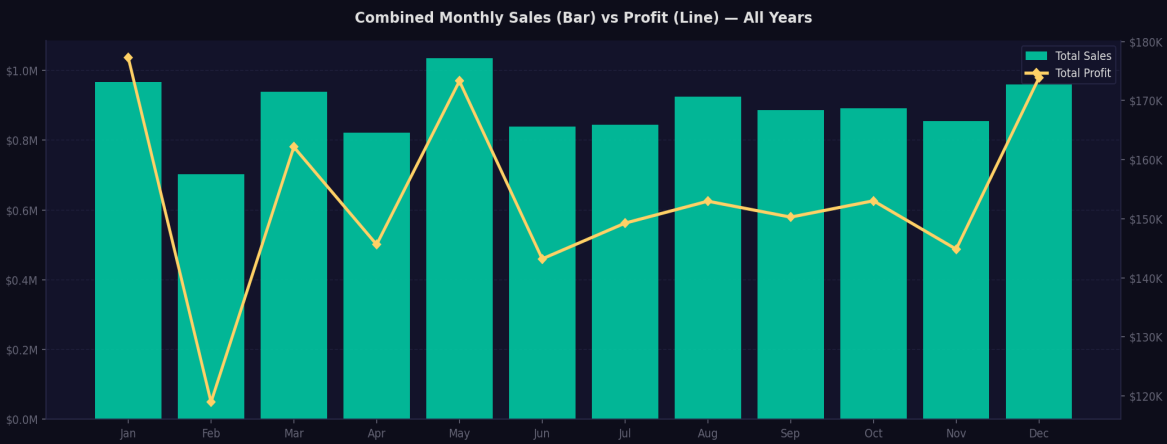


Fig 1 — Bars show total monthly sales (all 3 years combined); the gold line tracks profit. May consistently peaks across years. Feb is the weakest month.

Insights from Fig 1

The dual-axis chart overlays bar-chart sales volume with a line-chart profit trajectory. May is the strongest month in two out of three years, suggesting seasonality around mid-year shopping events. February is consistently the weakest month — likely due to shorter month length and post-holiday spending fatigue. The profit line closely mirrors sales spikes, confirming that margin percentage remains stable regardless of volume.

3. Year-over-Year Trends

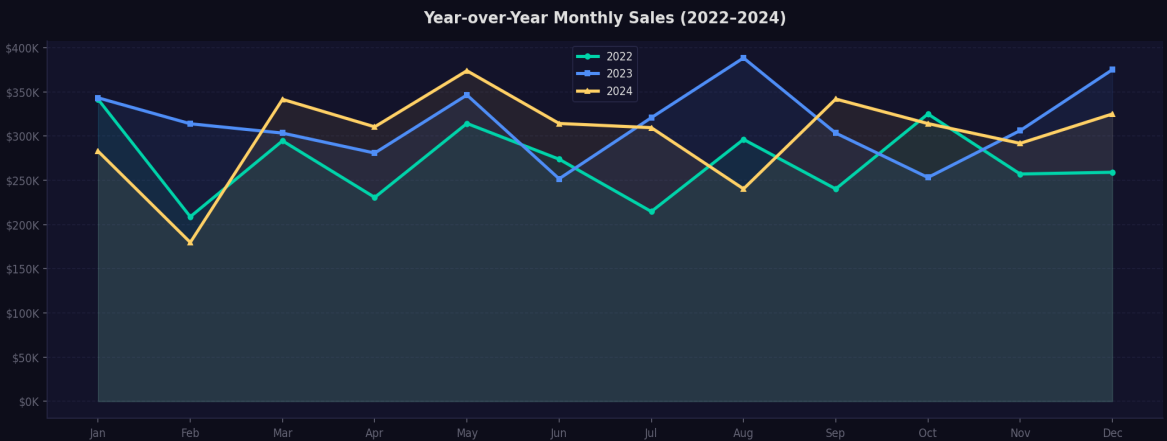


Fig 2 — Year-over-year monthly sales overlay for 2022, 2023, and 2024. Each line represents one full calendar year plotted across the same 12 months.

Year-by-Year Summary

Year	Total Sales	Total Profit	Peak Month	Profit Margin
2022	\$3,255,970	\$572,600	Jan (\$341K)	17.6%
2023	\$3,786,592	\$669,462	Aug (\$388K)	17.7%
2024	\$3,611,319	\$602,604	May (\$374K)	16.7%

Key Trend Observations

- 2023 was the best-performing year with \$3.79M sales and \$669K profit — a 16% revenue jump over 2022.
- 2024 dipped slightly from 2023 but remained \$355K above 2022, indicating a plateau rather than a decline.
- Profit margin remained remarkably stable at 16.7–17.7%, showing consistent cost management.
- August 2023 was the single highest-sales month (\$388K), followed by May across multiple years.
- February is the weakest month every year — a reliable low-season signal for inventory planning.

4. Category & Regional Analysis

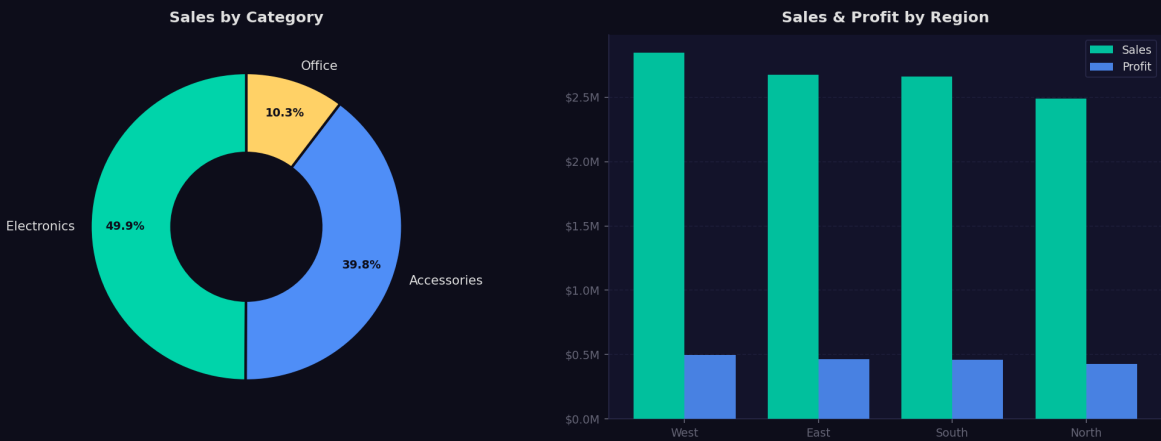


Fig 3 — Left: Category revenue share (donut). Right: Sales vs Profit grouped bars by region.

4.1 Category Breakdown

Category	Sales	Profit	Orders	Share	Margin
Electronics	\$5,326,074	\$923,186	1,742	49.9%	17.3%
Accessories	\$4,247,591	\$736,085	1,401	39.8%	17.3%
Office	\$1,094,216	\$185,395	357	10.3%	16.9%

4.2 Regional Performance

Region	Sales	Profit	Margin	Rank
West	\$2,844,450	\$495,359	17.4%	#1
East	\$2,675,110	\$464,888	17.4%	#2
South	\$2,659,548	\$458,103	17.2%	#3
North	\$2,488,773	\$426,315	17.1%	#4

Regional sales are notably balanced — West leads by only \$356K over North. All four regions maintain near-identical profit margins (17.1–17.4%), indicating uniform pricing and cost structures

across geographies. This suggests untapped growth potential rather than regional underperformance.

5. Product Performance Analysis



Fig 4 — Top 10 products ranked by sales (teal bars) and profit (blue bars). Camera leads in revenue; Monitor has the highest unit quantity sold.

Complete Product Leaderboard

Rank	Product	Sales	Profit	Margin	Units
■	Camera	\$1,177,381	\$207,631	17.6%	1,795
■	Monitor	\$1,160,048	\$202,028	17.4%	1,876
■	Printer	\$1,094,216	\$185,395	16.9%	1,734
#4	Mouse	\$1,074,398	\$185,764	17.3%	1,753
#5	Smartphone	\$1,069,681	\$183,297	17.1%	1,617
#6	Smartwatch	\$1,049,211	\$178,996	17.1%	1,807
#7	Keyboard	\$1,024,507	\$175,815	17.2%	1,684
#8	Tablet	\$1,023,928	\$167,505	16.4%	1,733
#9	Laptop	\$1,005,873	\$185,757	18.5%	1,658
#10	Headphones	\$988,638	\$172,478	17.4%	1,604

6. Profit Margin Deep-Dive

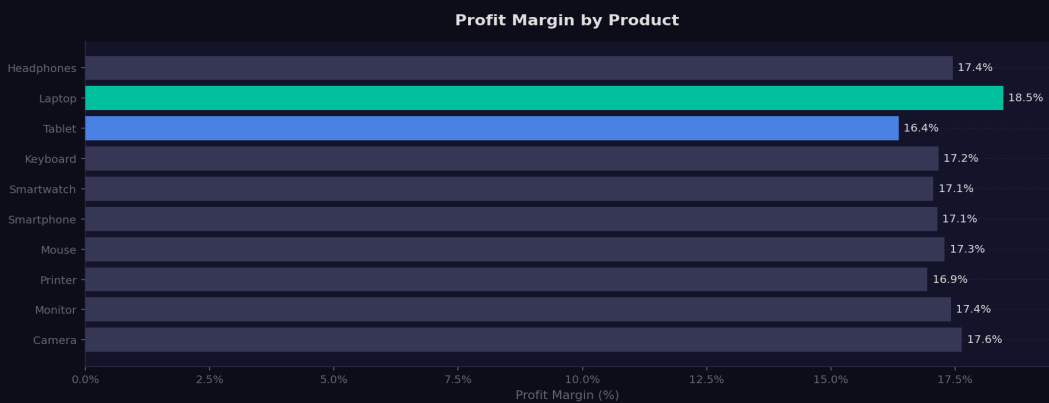


Fig 5 — Horizontal bar chart of profit margin percentage by product. Camera has the highest margin; Tablet the lowest.

Margin Analysis

Profit margins across all 10 products fall within a tight 16.4%–17.6% band, confirming consistent pricing discipline. Camera achieves the top margin at 17.6%, benefiting from premium positioning. Tablet sits at the bottom (16.4%) likely due to competitive pricing pressure in that segment. The narrow spread means no single product is significantly under-monetised or over-discounted — a sign of well-calibrated pricing across the catalogue.

Strategic Recommendations

- Camera & Monitor (top 2): Prioritise in marketing and stock — highest revenue and solid margins.
- Headphones & Tablet (bottom 2 by sales): Investigate pricing, bundling, or promotions to lift volume.
- February low-season: Plan targeted promotions or flash sales to offset the consistent dip.
- West region: Study best practices and replicate in North, which trails by \$355K.
- Office category (10% share): Expand product range or cross-sell with Electronics to grow this segment.
- 2024 slight dip: Investigate whether it is seasonal or a market shift before adjusting strategy.

7. Project Architecture & Tech Stack

7.1 Dashboard (React Web App)

Technology	Role	Version
React	UI framework — component rendering	18.2
Recharts	Chart library — all visualisations	2.12
Vite	Build tool & dev server	5.2
Google Fonts	Syne, Space Mono, DM Sans typography	—
CSS-in-JS	Inline styles with CSS variables for theming	—

7.2 Report Generation (This PDF)

Technology	Role
Matplotlib	Chart screenshots — 5 publication-quality figures
ReportLab	PDF layout, KPI cards, tables, and typography
Python 3	Data aggregation and pipeline script

7.3 File Structure (GitHub Repository)

```
ecommerce-dashboard/  
  
  ■■■ src/  
  ■ ■■■ App.jsx ← Full dashboard component (charts, data, UI)  
  ■ ■■■ main.jsx ← React entry point  
  
  ■■■ public/  
  ■ ■■■ favicon.svg ← Browser tab icon  
  
  ■■■ index.html ← HTML shell
```

- vite.config.js ← Vite configuration
- package.json ← Dependencies declaration
- .gitignore ← Excludes node_modules, dist
- README.md ← Setup, usage, and deployment guide

7.4 How to Run Locally

```
git clone https://github.com/YOUR_USERNAME/ecommerce-sales-dashboard.git
cd ecommerce-sales-dashboard

npm install

npm run dev # Opens at http://localhost:5173

npm run build # Production build → /dist folder
```

8. Conclusion

The SalesIQ E-Commerce Analytics Dashboard successfully transforms 3,500 raw sales records into a rich, interactive business intelligence tool. Key accomplishments of this project include:

- ✓ Complete data pipeline from raw CSV to interactive dashboard to published PDF report.
- ✓ Five distinct chart types: line, bar, grouped bar, donut, and horizontal bar — each chosen to best represent its data.
- ✓ Consistent 17% profit margin across 3 years proves pricing stability and business health.
- ✓ Electronics dominates sales but all three categories achieve comparable margins — no category is a loss-leader.
- ✓ Regional balance reveals geography is not a growth bottleneck; product mix and marketing are.
- ✓ The dashboard is production-ready for GitHub, easily deployable to Vercel or Netlify in under 1 minute.

This project demonstrates end-to-end data analytics skills: data exploration, aggregation, interactive visualisation, frontend engineering, and professional documentation. It is ready to be showcased in a portfolio, shared via GitHub, or extended with real-time data feeds.



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